

Abstract No.

BRD9 inhibition enhances radiotherapy response in glioblastoma by disrupting MYC-driven translational programs

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Radiotherapy (RT) remains a cornerstone of glioblastoma treatment; however, intrinsic and acquired radioresistance severely limit therapeutic efficacy and contribute to tumor recurrence. Adaptive transcriptional and translational programs activated after ionizing radiation (IR) enable tumor cells to survive genotoxic stress. Identifying molecular regulators that can prime GBM cells for enhanced radiation response represents a critical unmet need.

An epigenetic small-molecule library screen (125 compounds) combined with IR (4 Gy) identified bromodomain-containing protein 9 (BRD9) inhibition as a prominent radiosensitizing strategy. Pharmacological inhibition using structurally distinct BRD9 inhibitors and CRISPR/Cas9-mediated BRD9 loss significantly enhanced IR-induced clonogenic death in glioblastoma cells, while non-malignant astrocytes were minimally affected, indicating tumor-selective vulnerability. BRD9 inhibition primed glioblastoma cells for radiotherapy by increasing apoptotic fractions and delaying resolution of γ H2AX and 53BP1 foci following irradiation.

Transcriptomic profiling revealed marked downregulation of MYC target genes, ribosomal biogenesis pathways, and translational programs upon BRD9 perturbation. Consistently, global protein synthesis was reduced upon BRD9 inhibition, suggesting that BRD9 activity sustains MYC-dependent translational capacity required for radiation tolerance. Long-term IR-surviving tumor cell populations exhibited elevated BRD9 expression, supporting its role in adaptive radioresistance. Importantly, ectopic MYC overexpression restored clonogenic survival despite BRD9 inhibition, reversing the primed radiosensitive state.

These findings demonstrate that BRD9 inhibition primes glioblastoma cells for radiotherapy by suppressing MYC-driven translation and highlight a promising radiosensitization strategy.



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